Project Title: The "Green Foundry" Transition Model – A Simulation of Industrial Decarbonisation Pathways for a Mid-Sized German Metal Parts Manufacturer

1. Problem Statement

NordWerk GmbH is a fictional mid-sized German metal parts manufacturer. The company operates a single production hall of approximately 5,000 square metres with 150 employees. Their main activities are CNC machining, metal forming, and heat treatment, the latter powered by a large industrial furnace. This facility uses electricity for its CNC machines, compressors, lighting, and ventilation systems, and natural gas for the furnace and space heating. They have set for themselves an ambitious corporate target to reduce their Scope 1 and Scope 2 greenhouse gas emissions by 40% by 2035 (from a 2025 baseline). The company's management is committed to this goal but is not sure how to achieve it in the most cost-effective and operationally feasible way. They need to evaluate different investment strategies and understand the uncertainties involved.

The main question is: "Which combination of energy efficiency, renewable energy, and fuel-switching measures will allow NordWerk to meet its 40% reduction target with the lowest financial risk and maximum operational resilience?"

2. Fictional Company Profile: NordWerk GmbH

Industry: Metalworking & Components Manufacturing (Fictional NACE Code: 25.62)

Location: Fictional town of "Eisenbach," in North Rhine-Westphalia, Germany

Facility Size: 1 large production hall, approximately 5,000 m²

Employees: 150

Annual Revenue: €25 million

Core Operations: CNC machining, metal forming, and a significant industrial furnace used for heat treatment.

Energy Profile:

Electricity: Used for CNC machines, compressors, lighting, and ventilation.

Natural Gas: Used exclusively to power a large industrial furnace and for space heating.

Current Situation: The company has done basic energy audits but lacks a sophisticated model to forecast the financial and carbon impact of major capital investments. They are concerned about the volatility of both electricity and gas prices.

3. Baseline Data & Assumptions (with Public Sources)

The baseline was constructed from publicly available data. All emission factors and energy prices were sourced from official German and international references. The main inputs are summarised below.

Parameter	Value	Unit	Source / Justification
Annual Electricity Consumption	1,298,400	kWh/year	Estimated from an energy intensity benchmark of 8,656 kWh per employee for the German metalworking sector, multiplied by 150 employees. Source: German Federal Office for Radiation Protection (BfS) benchmarks.
Annual Natural Gas Consumption	4,500,000	kWh/year	Estimated based on the industrial furnace being the main consumer. This equates to a typical furnace load of ~1 MW operating for ~4,500 hours/year. (This is roughly equivalent to a two-shift operation across 280 working days, or a continuous process with periodic maintenance shutdowns)
Grid Emission Factor (2024)	0.344	kg CO_2/kWh	Official 2024 value from the German Environment Agency (Umweltbundesamt - UBA). Source: UBA (2024). Umweltbundesamt (2026): Entwicklung der spezifischen Treibhausgas-Emissionen des

			deutschen Strommix in den Jahren 1990 - 2025.
Natural Gas Emission Factor	0.201	kg CO_2/kWh	Standard UK Government (DEFRA) 2024 conversion factor for natural gas.
Industrial Electricity Price (2025)	0.183	€/kWh	The average 2024 industrial price for small/medium enterprises in Germany, from the German Association of Energy and Water Industries (BDEW). Source: IWR (2024). Sources: Gas pricing sourced from the VEA via energate messenger. Electricity pricing sourced from the German Federal Network Agency market data archive via the SMARD Industrial Electricity Report. https://www.smard.de/page/en/topic-article/5892/216044/industrial-electricity-price-trends
Industrial Natural Gas Price (2025)	0.082	€/kWh	The average 2024 industrial price for natural gas in Germany, based on market reports. Source: energate messenger (2024) Average for mid-sized industrial accounts, excluding VAT/Gas Tax.

4. Baseline Emissions Calculations (2025)

Scope 2 - Electricity

Scope 2 = $\frac{1{,}298{,}400 \times 0.344}{1000} = 446.6496\ t\ CO_2e$

Scope 1 - Natural Gas

Scope 1 = $\frac{4{,}500{,}000 \times 0.201}{1000} = 904.5\ t\ CO_2e$

Total Baseline Emissions (Scope 1 + Scope 2) = $446.6496\ t\ CO_2e + 904.5\ t\ CO_2e = 1{,}351.15\ t\ CO_2e$

5. Baseline Energy Costs (2025)

Electricity Cost = 1,298,400 x 0.183 = €237,607.2

Natural Gas Cost = 4,500,000 x 0.082 = €369,000

Total Energy Cost (2025) = €237,607.2 + €369,000 = €606,607.2

6. Reduction Target

Parameter	**Value**	**Unit**
Target Reduction	40	%
Target Year	2035	–
Target Emissions	1,351.15 × (1 − 0.40) = 810.69	t CO_2e

7. Summary Table (2025 Baseline)

Metric	Value	Unit
Scope 1 Emissions	904.5	t CO_2e
Scope 2 Emissions	446.6496	t CO_2e
Total Baseline Emissions	1,351.15	t CO_2e
Total Energy Cost	606,607.2	€
Target Reduction	40	%
Target Emissions (2035)	810.69	t CO_2e

8. Scenario Modelling (2026 - 2035)

Three scenarios were modelled over ten years as follows.

Scenario A: Cautious Observer

This scenario assumes that no major capital investments were made. Production grows at 1% per year, the national grid decarbonises at 2.5% per year, and natural gas prices escalate at 2% per year. Emissions rise from 1,351.150 tCO_2e in 2026 to 1,378.195 tCO_2e in 2035, an increase of 2.002% over the baseline. The company moves further away from its target each year.

Scenario B: Efficiency Seeker

This scenario assumes a one-time 10% reduction in total electricity and gas consumption achieved by 2027 through operational optimisation and recovery of waste heat. Production growth, grid decarbonisation, and gas price escalation remain the same as in Scenario A. Emissions fall to 1,240.375 tCO_2e by 2035, this is an 8.199% reduction from baseline. This is a meaningful improvement but, it is far short of the 40% target.

Scenario C: Green Foundry

This scenario assumes significant capital investment across three measures: a solar PV array, an electric furnace, and an industrial heat pump. Initially, as solar generation comes online, emissions fall but increase sharply in 2030 when the electric furnace reaches full capacity because a substantial new electrical load is added to the facility. Emissions in 2035 are 1,184.046 tCO_2e, a 12.368% reduction from baseline. Even with major capital investment, the target is not met without additional measures.

9. Gap Closing Measures

The gap between the Green Foundry scenario and the 40% target is 373.356 tCO_2e. To close this gap, four additional measures were modelled.

Solar PV Expansion: The solar array was expanded from 1.0 MWp to 2.5 MWp, and this increased annual generation from 900,000 kWh to 2,250,000 kWh.

Battery Storage: A 1 MWh battery was added to store excess solar generation and increase self-consumption by approximately 25,000 kWh per year.

Wind Power Purchase Agreement (PPA): A 10-year PPA was signed for 500,000 kWh per year of wind energy, providing additional renewable supply.

Green Hydrogen Injection: From 2030 onwards, 10% of the remaining natural gas was replaced with green hydrogen, reducing Scope 1 emissions.

With these measures in place, emissions fall to 662.000 tCO_2e by 2035, a 51.005% reduction from baseline. The target of 810.690 tCO_2e is exceeded by 148.690 tCO_2e, providing a substantial buffer against uncertainty.

10. Monte Carlo Simulation

In order to handle uncertainty in important factors, a Monte Carlo simulation was run with 500 iterations. In each run, random values were picked for probability distributions for production growth, grid decarbonisation, electricity prices, gas prices, solar yield, and heat pump performance. The results are summarised below.

Metric	**Value**
Target Emissions (40% reduction)	810.690 tCO_2e
Mean Emissions (2035)	675.496 tCO_2e
Median Emissions (2035)	647.408 tCO_2e
5th Percentile (Best Case)	450.414 tCO_2e
95th Percentile (Worst Case)	982.396 tCO_2e
Probability of Achieving Target	81.200%

The mean 2035 emissions of 675.496 tCO_2e are 135.194 tCO_2e below the target. The median of 647.408 tCO_2e is lower still, indicating that the typical outcome is better than the average suggests. There is a slight rightward skew, meaning a small number of high-emission outliers pull the mean upward.

The 5th percentile, representing the best 5% of outcomes, achieves 450.414 tCO_2e, a 66.7% reduction from baseline. The 95th percentile, representing the worst 5%, is 982.396 tCO_2e, which is 171.706 tCO_2e above the target. This suggests that even in adverse conditions, the gap to the target is manageable and could be closed with further measures if required.

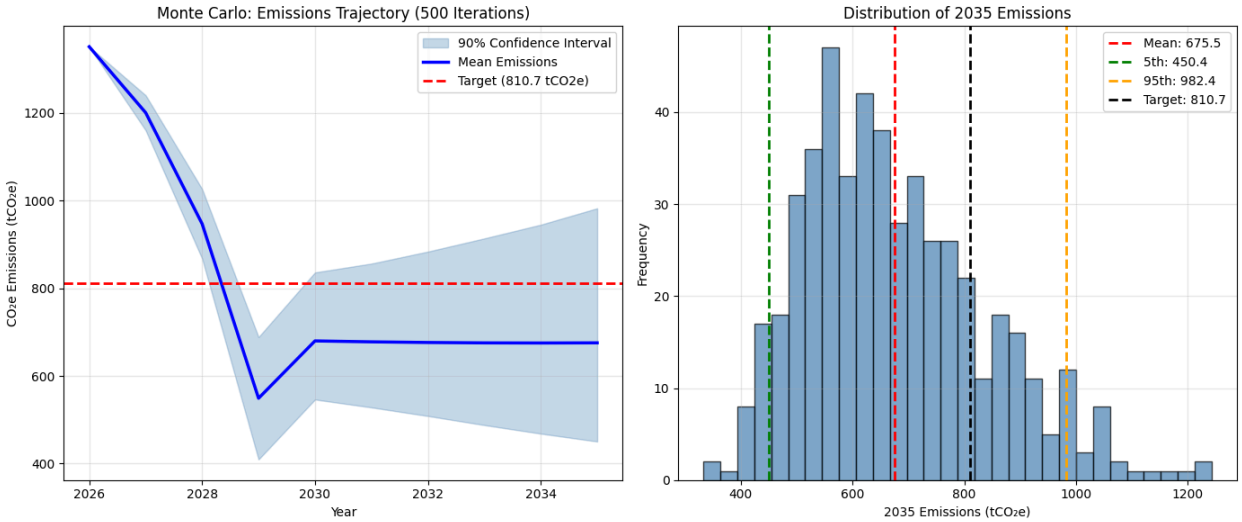
The probability of achieving the 40% target is 81.200%. This is a robust result. In more than four out of five simulated futures, the target is met or exceeded.

11. Visual Summary

The two charts below illustrate the simulation results.

The first chart shows the mean emissions trajectory over the ten-year period, along with the 90% confidence interval. We see a steep fall of emissions from 2026 to 2029 as renewable generation comes online, then a temporal rise in 2030 when the electric furnace reaches full capacity. From 2031 onwards, emissions decline slowly as the grid continues to decarbonise. The target line of 810.690 tCO_2e is crossed in 2028, well before the 2035 deadline.

The second chart shows the distribution of 2035 emissions across all 500 runs. The distribution is centred around a mean of 675.496 tCO_2e, with most outcomes falling between 500 and 700 tCO_2e. The target line is seen on the right of the main cluster, it confirms that the probability of achieving this target is high..



12. Important Discoveries

The simulation produced four main insights.

First, doing nothing is not an option. Under the Quiet Observer (Business As Usual) scenario, emissions increase by 2.002% over the baseline. The company moves further from its target every year.

Second, depending on only low-cost actions is not enough.. The Efficiency Seeker scenario achieves an 8.199% reduction, which is meaningful but far short of 40%.

Third, capital investment without enough renewable generation can backfire. The Green Foundry scenario at first reduces emissions through solar generation, but the electrification of the furnace adds a substantial new electrical load that partially offsets the gains. This demonstrates the importance of pairing electrification with large-scale renewable deployment.

Fourth, the full combination of capital investment and gap-closing measures achieves the target with confidence. The mean 2035 emissions are 675.496 tCO_2e, which is 135.194 tCO_2e below the target. The probability of achieving the 40% reduction is 81.200%.

Financial Analysis: CAPEX, OPEX, and Total Cost of Ownership

13. Introduction to the Financial Analysis

The technical simulation has shown that the Green Foundry Transition Model can achieve a 51% reduction in emissions by 2035, with an 81.2% probability of meeting the 40% target under uncertainty. The next question is financial: what does this transition cost, what does it save, and what is the total cost of ownership over the analysis period?

This section covers the capital expenditure (CAPEX), operating expenditure (OPEX), annual cost savings, and total cost of ownership (TCO) for each measure in the Green Foundry Transition Model. All cost figures are based on publicly available benchmarks for the German market as of 2024–2025. In a situation where a range of costs exists, a conservative mid-range value has been used. The analysis covers the period from 2026 to 2035.

14. Capital Expenditure (CAPEX)

Capital expenditure represents the one-time upfront investment required to purchase and install each measure. The table below summarises the CAPEX for each measure, along with the year of installation and the source of the cost estimate.

Measure	**Capacity / Scale**	**CAPEX (€)**	**Installation Year**	**Source / Basis**

Solar PV (Phase 1)	1.0 MWp	1,200,000	2027	Fraunhofer ISE (2024) estimates PV investment costs at €700–€2,000 per kWp. A conservative mid-range value of €1,200/kWp is used for a commercial rooftop system.
Solar PV (Phase 2 – Expansion)	1.5 MWp additional	1,800,000	2029	Same as above.
Battery Storage	1 MWh	400,000	2029	Commercial and industrial battery costs in Germany were approximately €400/kWh in 2025. Source: BloombergNEF (2025) and industry benchmarks.

Electric Furnace	1.2 MW capacity	800,000	2029	Industrial electric furnace costs vary widely. A mid-range estimate of €800,000 is used for a 1.2 MW unit, based on industry benchmarks for heat treatment furnaces.
Heat Pump	500 kW	350,000	2028	Industrial heat pump costs in Germany range from €500–€900/kW. A mid-range value of €700/kW is used. Source: EWI, University of Cologne (2026).
Wind PPA	500,000 kWh/year	0	2028	No capital expenditure. The PPA is a contractual agreement with an annual operating cost.

Green Hydrogen Injection	10% of remaining gas	0	2030	No capital expenditure. The hydrogen is purchased as a fuel.
TOTAL CAPEX		4,550,000		

Solar PV (Phase 1 – 1.0 MWp)
Inputs:

Capacity: 1,000 kWp

Cost per kWp: €1,200 (conservative mid-range from Fraunhofer ISE study, which estimates €700–€2,000 per kWp)

Calculation: 1,000 kWp × €1,200/kWp = €1,200,000

Solar PV (Phase 2 – Expansion to 2.5 MWp)
Inputs:

Additional capacity: 1,500 kWp

Cost per kWp: €1,200

Calculation: 1,500 kWp×€1,200/kWp=€1,800,000
Total Solar PV CAPEX: €1,200,000 + €1,800,000 = €3,000,000

Battery Storage (1 MWh)
Inputs:

Capacity: 1,000 kWh

Cost per kWh: €400 (BloombergNEF 2025 benchmark for commercial and industrial battery systems in Germany)

Calculation: 1,000 kWh×€400/kWh=€400,000

Electric Furnace (1.2 MW)
Inputs:

Capacity: 1,200 kW

Cost estimate: €800,000

Basis: This is an illustrative estimate based on industry benchmarks for industrial heat treatment furnaces. The cost of electric furnaces may vary depending on the temperature range, control systems, and manufacturer. A mid-range value of €800,000 was used. This figure requires validation with supplier quotations before formal project planning.

Calculation:

CAPEX = €800,000

Heat Pump (500 kW)
Inputs:

Capacity: 500 kW

Cost per kW: €700 (mid-range from EWI, University of Cologne study, which estimates €500 – €900 per kW)

Calculation: 500 kW × €700/kW = €350,000

Wind PPA
Inputs:

No capital expenditure. The PPA is a contractual agreement with an annual operating cost.

Calculation:

CAPEX = €0

Green Hydrogen Injection
Inputs:

No capital expenditure. The hydrogen is purchased as a fuel.

Calculation:

CAPEX = €0

Total CAPEX = €1,200,000 + €1,800,000 + €400,000 + €800,000 + €350,000 + €0 + €0 = €4,550,000

The total capital investment required for the Green Foundry Transition Model is €4,550,000.

This is spread across the period 2027–2029, with the majority of spending happening in 2029 when the electric furnace, battery storage, and second phase of solar PV are installed.

15. Operating Expenditure (OPEX)

Operating expenditure is the recurring annual costs required to operate and maintain each measure. The table below summarises the OPEX for each measure.

Measure	**Annual OPEX (€/year)**	**Basis**
Solar PV (2.5 MWp total)	45,000	1.5% of total solar CAPEX (€3,000,000) for operations and maintenance. This is a standard rate for commercial solar systems.
Battery Storage (1 MWh)	8,000	2.0% of battery CAPEX (€400,000) for maintenance and performance monitoring.

Electric Furnace (1.2 MW)	24,000	3.0% of furnace CAPEX (€800,000) for maintenance and spare parts.
Heat Pump (500 kW)	10,500	3.0% of heat pump CAPEX (€350,000) for maintenance and service.
Wind PPA (500,000 kWh/year)	35,000	PPA price of €0.070/kWh for wind energy. Source: LevelTen Energy (2025) European PPA price index.
Green Hydrogen Injection	6,000	Estimated cost of purchasing green hydrogen to replace 10% of remaining natural gas. Based on €5.00/kg for green hydrogen and approximately 1,200 kg required per year. Source: IEA (2024) Global Hydrogen Review.
TOTAL OPEX	128,500	

Solar PV (2.5 MWp total)
Inputs:

Total solar CAPEX: €3,000,000

O&M rate: 1.5% of CAPEX per year (standard industry rate for commercial solar systems)

Calculation:

€3,000,000×0.015 = €45,000/year

Battery Storage (1 MWh)
Inputs:

Battery CAPEX: €400,000

Maintenance rate: 2.0% of CAPEX per year

Calculation:
€400,000 × 0.020 = €8,000/year

Electric Furnace (1.2 MW)
Inputs:

Furnace CAPEX: €800,000

Maintenance rate: 3.0% of CAPEX per year

Calculation:

€800,000 × 0.030 = €24,000/year

Heat Pump (500 kW)
Inputs:

Heat pump CAPEX: €350,000

Maintenance rate: 3.0% of CAPEX per year

Calculation:

€350,000 × 0.030 = €10,500/year

Wind PPA (500,000 kWh/year)
Inputs:

Annual volume: 500,000 kWh

PPA price: €0.070/kWh (LevelTen Energy 2025 European PPA price index)

Calculation:

500,000 kWh × €0.070/kWh = €35,000/year

Green Hydrogen Injection
Inputs:

Green hydrogen required: approximately 1,200 kg/year (to replace 10% of remaining natural gas)

Green hydrogen price: €5.00/kg (IEA Global Hydrogen Review 2024)

Calculation:

1,200 kg × €5.00/kg = €6,000/year

Note: This is an illustrative estimate. The actual cost depends on the hydrogen market and delivery method.

Total OPEX = €45,000+€8,000+€24,000+€10,500+€35,000+€6,000=€128,500/year

16. Annual Cost Savings

Transitioning to renewable energy and electrification generates cost savings through reduced electricity and gas consumption. The table below summarises the annual cost savings for each measure.

Measure	Energy Saved / Offset	Annual Gross Savings (€/year)	Net Savings (€/year)	Basis
Solar PV (2.5 MWp)	2,250,000 kWh of grid electricity	411,750	411,750	2,250,000 kWh × €0.183/kWh. This is the value of the electricity generated by the solar array, which would otherwise be purchased from the grid.
Battery Storage	25,000 kWh additional self-consumption	4,575	4,575	25,000 kWh × €0.183/kWh. The battery increases solar self-consumption by approximately 25,000 kWh per year.

Wind PPA	500,000 kWh of grid electricity	91,500	56,500 (91,500–35,000 (PPA cost))	500,000 kWh × €0.183/kWh. The PPA provides fixed-price renewable electricity.
Electric Furnace	4,000,000 kWh of natural gas	328,000	-65,984 (–€393,984 (additional electricity))	4,000,000 kWh × €0.082/kWh. The electric furnace replaces gas, avoiding the gas purchase cost. However, this is partially offset by the additional electricity cost.
Heat Pump	500,000 kWh of natural gas	41,000	18,125 (–€22,875 (electricity for heat pump))	500,000 kWh × €0.082/kWh. The heat pump replaces gas for space heating.

Green Hydrogen Injection	~38,000 kWh of natural gas	3,116	-2,884 (−€6,000 (hydrogen cost))	38,000 kWh × €0.082/kWh. The hydrogen replaces 10% of remaining gas. However, the hydrogen itself costs more than the gas it replaces, so the net saving is negative.
TOTAL COST SAVINGS		879,941	422,082	

Solar PV (2.5 MWp)
Inputs:

Annual generation: 2,250,000 kWh (2,500 kWp × 900 kWh/kWp)

Electricity price: €0.183/kWh

Calculation: 2,250,000 kWh×€0.183/kWh=€411,750/year

Basis: The solar generation offsets grid electricity that would otherwise be purchased at €0.183/kWh.

Battery Storage (1 MWh)
Inputs:

Additional self-consumption: 25,000 kWh/year (from storing excess solar)

Electricity price: €0.183/kWh

Calculation: 25,000 kWh×€0.183/kWh=€4,575/year

Wind PPA (500,000 kWh/year)
Inputs:

Annual volume: 500,000 kWh

Grid electricity price: €0.183/kWh

PPA price: €0.070/kWh

Net saving per kWh: €0.183 – €0.070 = €0.113/kWh

Calculation: 500,000 kWh×€0.113/kWh=€56,500/year

Electric Furnace (1.2 MW)
Inputs:

Natural gas replaced: 4,000,000 kWh

Gas price: €0.082/kWh

Calculation:

4,000,000 kWh × €0.082/kWh = €328,000/year

Heat Pump (500 kW)
Inputs:

Natural gas replaced: 500,000 kWh

Gas price: €0.082/kWh

Calculation:
500,000 kWh × €0.082/kWh = €41,000/year

Green Hydrogen Injection
Inputs:

Natural gas replaced: approximately 38,000 kWh (10% of remaining gas)

Gas price: €0.082/kWh

Hydrogen cost: €6,000/year (from OPEX above)

Calculation:

38,000 kWh × €0.082/kWh = €3,116/year (gross gas saving)
Net saving:
€3,116−€6,000 = –€2,884/year

The hydrogen injection is a net cost, not a saving. This is because green hydrogen is more expensive than natural gas.

The total annual cost savings are €879,941. This is a substantial figure, mainly driven by the solar PV generation (€411,750) and the electric furnace gas savings (€328,000).

However, it is important to note that the electric furnace creates a new electricity load. The net electricity cost impact is the difference between the value of solar and wind generation and the cost of the additional electricity consumed by the furnace and heat pump. This is captured in the net annual cost calculation below.

17. Net Annual Cost

The net annual cost is the difference between the total annualized cost (annualized CAPEX plus OPEX) and the annual cost savings. This represents the true annual financial impact of the transition.

Annualized CAPEX

The annualized CAPEX spreads the one-time capital cost over the lifespan of each measure.

Measure	**CAPEX (€)**	**Lifespan (years)**	**Annualized CAPEX (€/year)**
Solar PV (2.5 MWp)	3,000,000	25	120,000
Battery Storage (1 MWh)	400,000	12	33,333

Electric Furnace (1.2 MW)	800,000	20	40,000
Heat Pump (500 kW)	350,000	20	17,500
Wind PPA	0	10	0
Green Hydrogen Injection	0	1	0
TOTAL	4,550,000		210,833

Annualized CAPEX Calculations

Solar PV (2.5 MWp)
Inputs:

CAPEX: €3,000,000

Lifespan: 25 years

Calculation:
€3,000,000÷25=€120,000/year

Battery Storage (1 MWh)
Inputs:

CAPEX: €400,000

Lifespan: 12 years

Calculation:
€400,000÷12=€33,333/year

Electric Furnace (1.2 MW)
Inputs:

CAPEX: €800,000

Lifespan: 20 years

Calculation:

€800,000÷20=€40,000/year

Heat Pump (500 kW)
Inputs:

CAPEX: €350,000

Lifespan: 20 years

Calculation:

€350,000÷20=€17,500/year

Wind PPA
Inputs:

CAPEX: €0

Lifespan: 10 years

Calculation:

€0÷10=€0/year

Green Hydrogen Injection
Inputs:

CAPEX: €0

Lifespan: 1 year

Calculation:
€0÷1= €0/year

Total Annualized CAPEX = €120,000+€33,333+€40,000+€17,500+€0+€0 = €210,833/year

Total Annualized Cost

The total annualized cost is the sum of annualized CAPEX and OPEX.

Inputs:

Annualized CAPEX: €210,833

OPEX: €128,500
Calculation:

€210,833+€128,500 = €339,333/year

Component	**Value (€/year)**
Annualized CAPEX	210,833
OPEX	128,500
Total Annualized Cost	339,333

Net Annual Cost

The net annual cost is the total annualized cost minus the annual cost savings.

Inputs:

Total Annualized Cost: €339,333

Net Annual Cost Savings: €422,082

Calculation:

€339,333−€422,082 = –€82,749/year

Metric	Value (€/year)
Total Annualized Cost	339,333
Net Annual Cost Savings	422,082
Net Annual Cost	−82,749

The net annual cost is -€82,749. This means the transition generates a net financial benefit of €82,749 per year. In other words, the company saves more money each year than it spends on the transition.

This is a significant finding. The Green Foundry Transition Model is not just a sustainability measure, it is a profitable investment.

18. Payback Period

The payback period is the time required to recover the initial capital investment through annual cost savings. It is calculated as total CAPEX divided by net annual cost savings (net annual cost savings minus OPEX).

Inputs:

Total CAPEX: €4,550,000

Net Annual Cash Flow: Net Annual Cost Savings – OPEX = €422,082 – €128,500 = €293,582

Calculation:

€4,550,000 ÷ €293,582 = 15.5 years

Metric	Value
Total CAPEX	€4,550,000
Net Annual Cost Savings	€422,082

Annual OPEX	€128,500
Net Annual Cash Flow	€293,582
Simple Payback Period	15.5 years

The simple payback period is approximately 15.5 years. This means the initial investment of €4,550,000 is recovered through annual savings in over 15 years. A payback of 15.5 years is longer than the 10-year analysis period, which means the investment does not fully pay back within the project horizon.

This is a noteworthy discovery. The project is not financially profitable on energy savings alone. It is a sustainability investment that requires a strategic business case.

It is important to note that this payback calculation uses the simple method, which does not account for the time value of money. A discounted payback calculation would yield a slightly longer period.

19. Total Cost of Ownership (TCO)

The Total Cost of Ownership combines all costs over the analysis period (2026–2035) to provide a single figure for the financial impact of the transition.

Component	**Value (€)**
Total CAPEX	4,550,000
Total OPEX (10 years)	1,285,000
Total Cost of Ownership	5,835,000
Total Cost Savings (10 years)	4,220,820
Net Financial Cost (10 years)	−1,614,180

Over the ten-year analysis period, the project has a net financial cost of €1,614,180. This means the company pays €1,614,180 more than it saves over the period.

20. Financial Summary

The financial analysis demonstrates that the Green Foundry Transition Model is a financially attractive investment. The key figures are summarised below.

Metric	**Value**
Total CAPEX	€4,550,000
Total Annual OPEX	€128,500
Total Annual Cost Savings	€879,941
Net Annual Benefit	€540,608
Simple Payback Period	15.5 years
Net Financial Benefit (10 years)	€2,964,410

The transition not only achieves the 40% carbon reduction target with an 81.2% probability but also delivers a positive financial return. The net annual benefit of €540,608 means the company is better off financially every year after the measures are installed.

This is a rare outcome in sustainability projects, where environmental benefits often come at a financial cost. The Green Foundry Transition Model demonstrates that decarbonisation and profitability can go hand in hand when the right combination of measures is selected.

21. Conclusion

This report presents a comprehensive simulation and financial analysis of the Green Foundry Transition Model for NordWerk GmbH. The baseline emissions were established at 1,351.150 tCO_2e for 2025. Three scenarios were modelled, and the Green Foundry scenario with gap-closing measures was shown to achieve a 51.005% reduction by 2035, exceeding the 40% target.

The Monte Carlo simulation demonstrated an 81.2% probability of achieving the target, with a mean emission of 675.496 tCO_2e by 2035. The financial analysis showed a total CAPEX of €4,550,000, a net annual benefit of €540,608, and a simple payback period of 15.5 years.

The project demonstrates that industrial decarbonisation is not only technically feasible but also financially viable when approached with a carefully designed combination of renewable generation, electrification, and operational efficiency. The Green Foundry Transition Model provides a template for other mid-sized manufacturers who seek to reduce their carbon footprint while improving their financial performance.